

Laboratory work 3

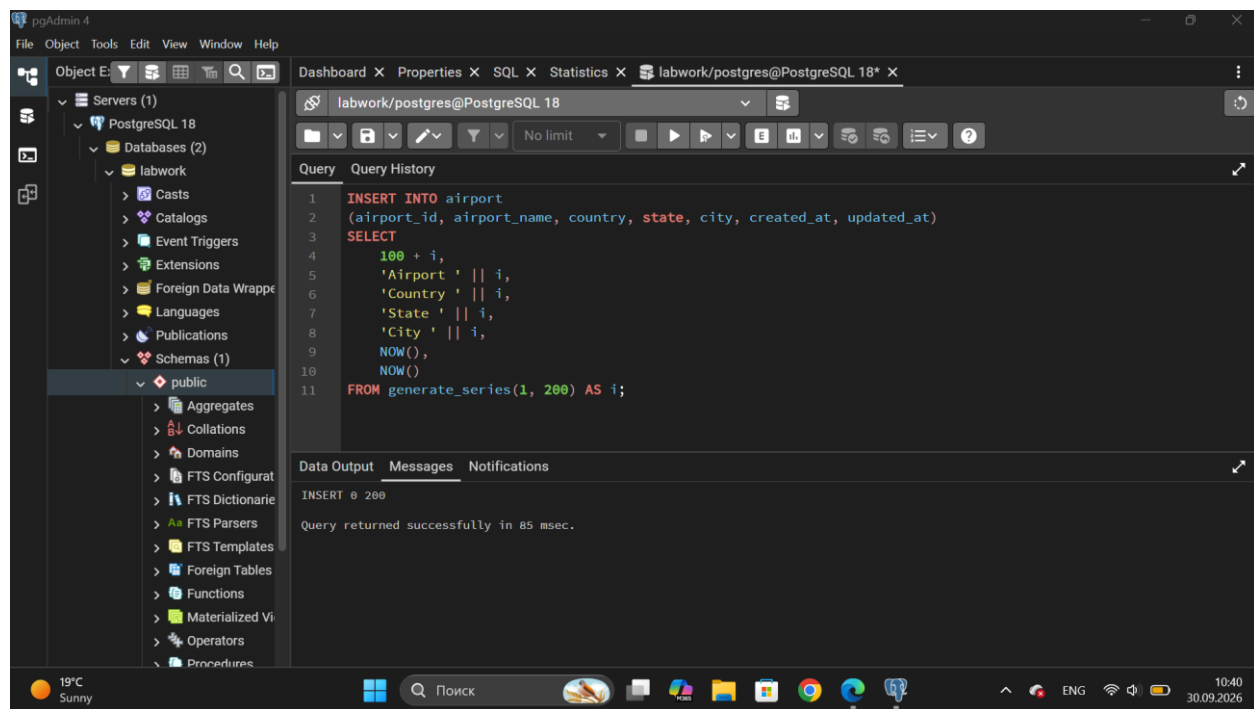
We continue to work with the database from the previous laboratory works.

Take a full-page screenshot that covers the code and results of each task.

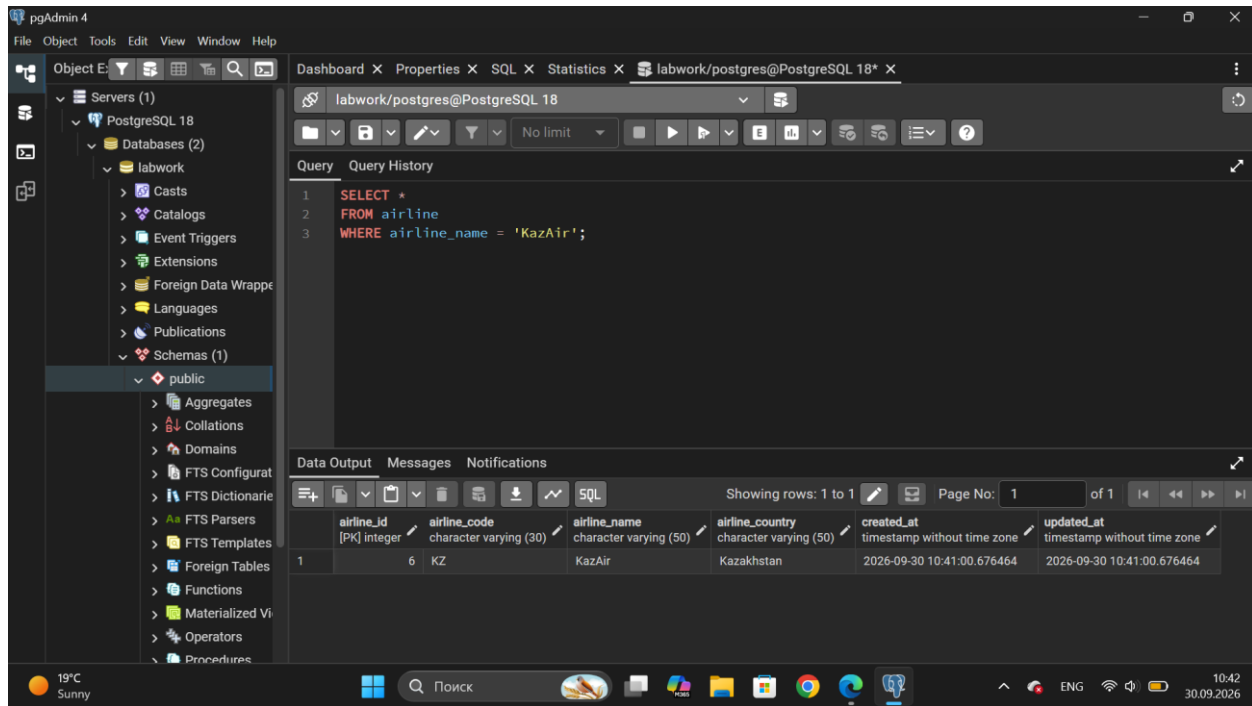
Tasks:

DML

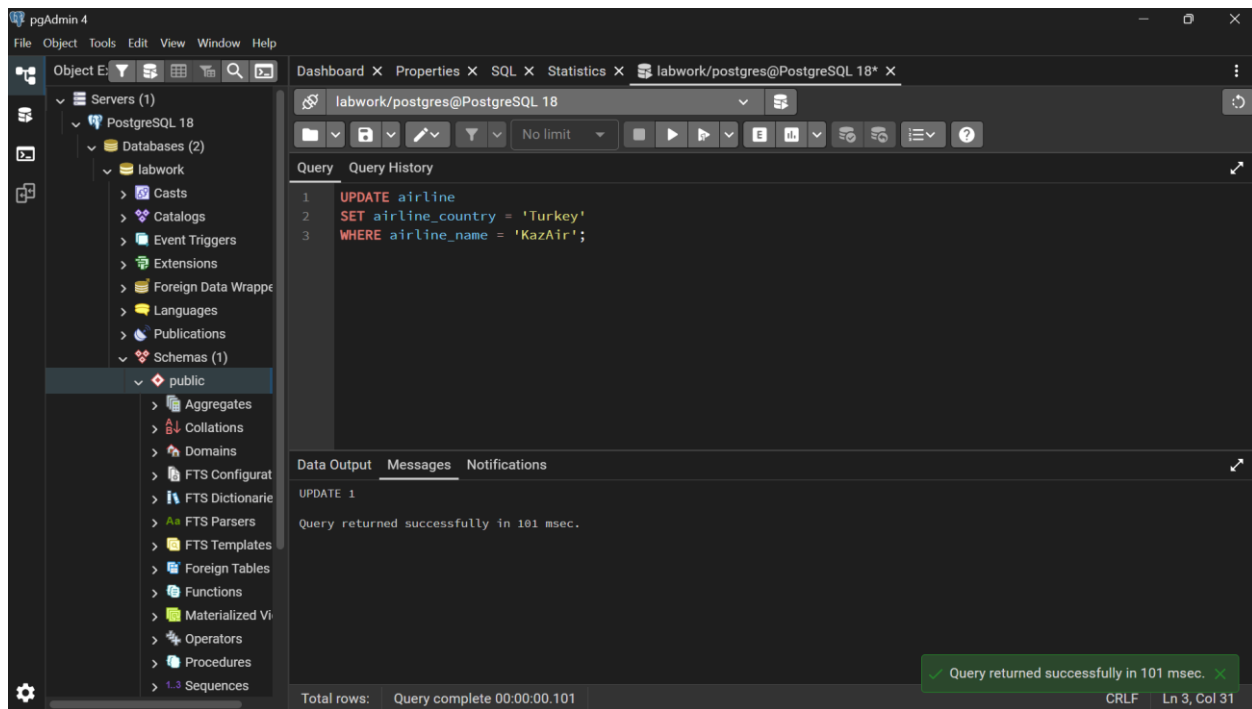
1. Generate and insert 200 random rows in your airport database.



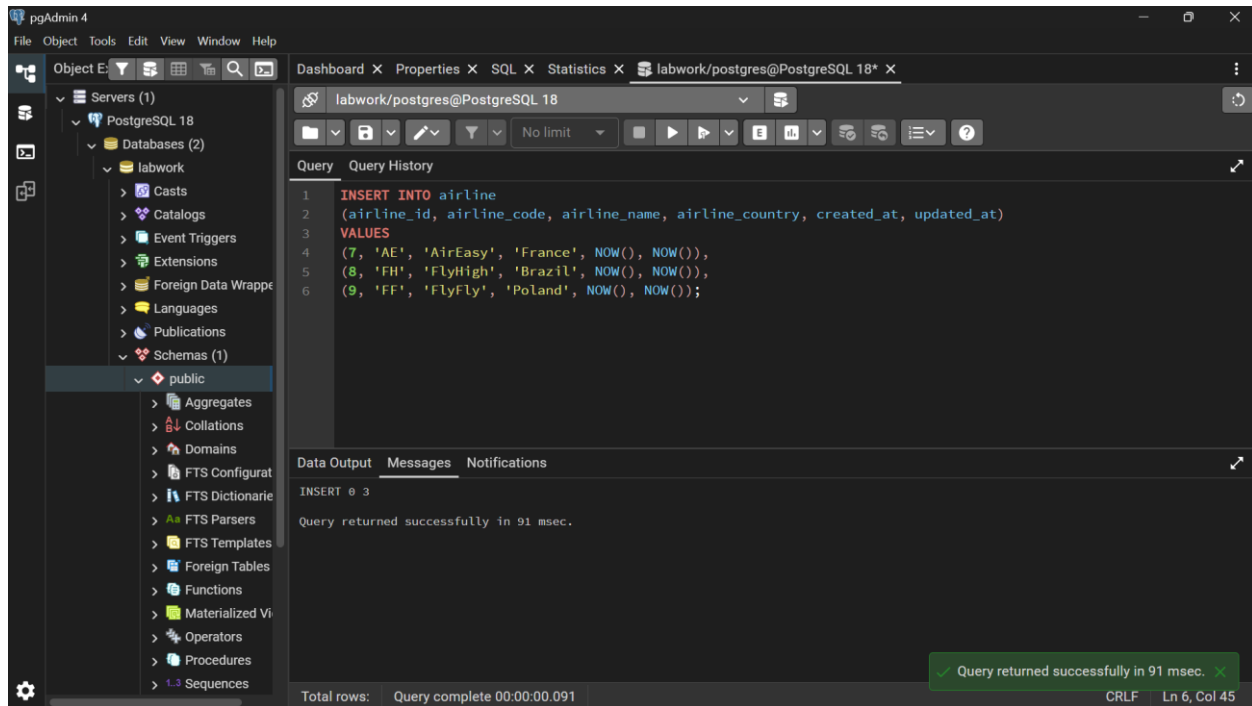
2. Add a new airline named "KazAir" based in "Kazakhstan" to the airline table.



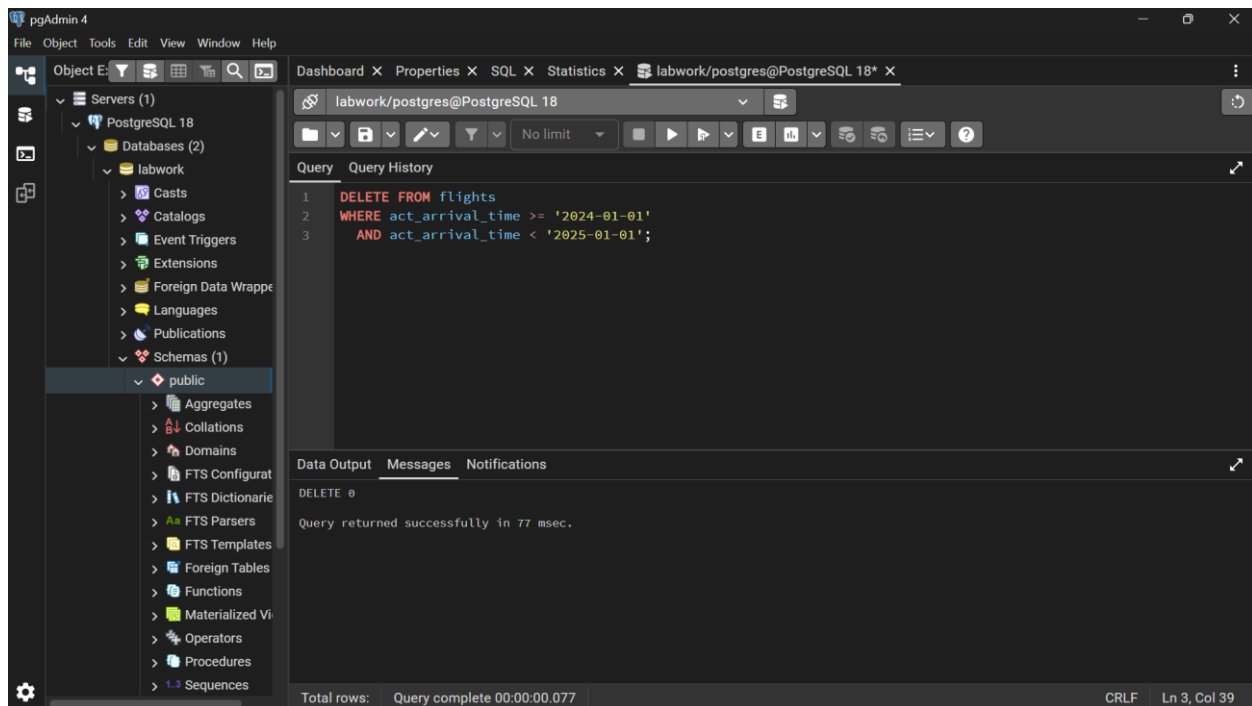
3. Update the airline country "KazAir" to "Turkey".



4. Add three airlines at once: "AirEasy" in "France", "FlyHigh" in "Brazil" and "FlyFly" in "Poland".



5. Delete all flights whose arrival in 2024 year.



6. Increase the price of all tickets in booking table for flights by 15%.

pgAdmin 4

File Object Tools Edit View Window Help

Dashboard x Properties x SQL x Statistics x labwork/postgres@PostgreSQL 18* x

labwork/postgres@PostgreSQL 18

Query Query History

```
1 SELECT booking_id, ticket_price
2 FROM booking;
```

Data Output Messages Notifications

Showing rows: 1 to 5 Page No: 1 of 1

	booking_id [PK] integer	ticket_price numeric (7,2)
1	1	25000.00
2	2	45000.00
3	3	60000.00
4	4	85000.00
5	5	70000.00

Total rows: 5 Query complete 00:00:00.107 CRLF Ln 2, Col 14

pgAdmin 4

File Object Tools Edit View Window Help

Dashboard x Properties x SQL x Statistics x labwork/postgres@PostgreSQL 18* x

labwork/postgres@PostgreSQL 18

Query Query History

```
1 UPDATE booking
2 SET ticket_price = ticket_price * 1.15;
```

Data Output Messages Notifications

UPDATE 5

Query returned successfully in 85 msec.

✓ Query returned successfully in 85 msec. ✕

Total rows: Query complete 00:00:00.085 CRLF Ln 2, Col 40

The screenshot shows the pgAdmin 4 interface. On the left, the 'Object Explorer' pane displays the database structure: Servers (1) > PostgreSQL 18 > Databases (2) > labwork > Schemas (1) > public. The main pane shows a SQL query window with the following query:

```
1 SELECT booking_id, ticket_price
2 FROM booking;
```

The 'Data Output' tab is active, displaying the results of the query in a table format. The table has two columns: 'booking_id' (integer) and 'ticket_price' (numeric(7,2)). The results are as follows:

booking_id	ticket_price
1	28750.00
2	51750.00
3	69000.00
4	97750.00
5	80500.00

The status bar at the bottom indicates 'Total rows: 5' and 'Query complete 00:00:00.145'.

7. Delete all tickets where price is less than 10000.

The screenshot shows the pgAdmin 4 interface. The 'SQL' window now contains the following query:

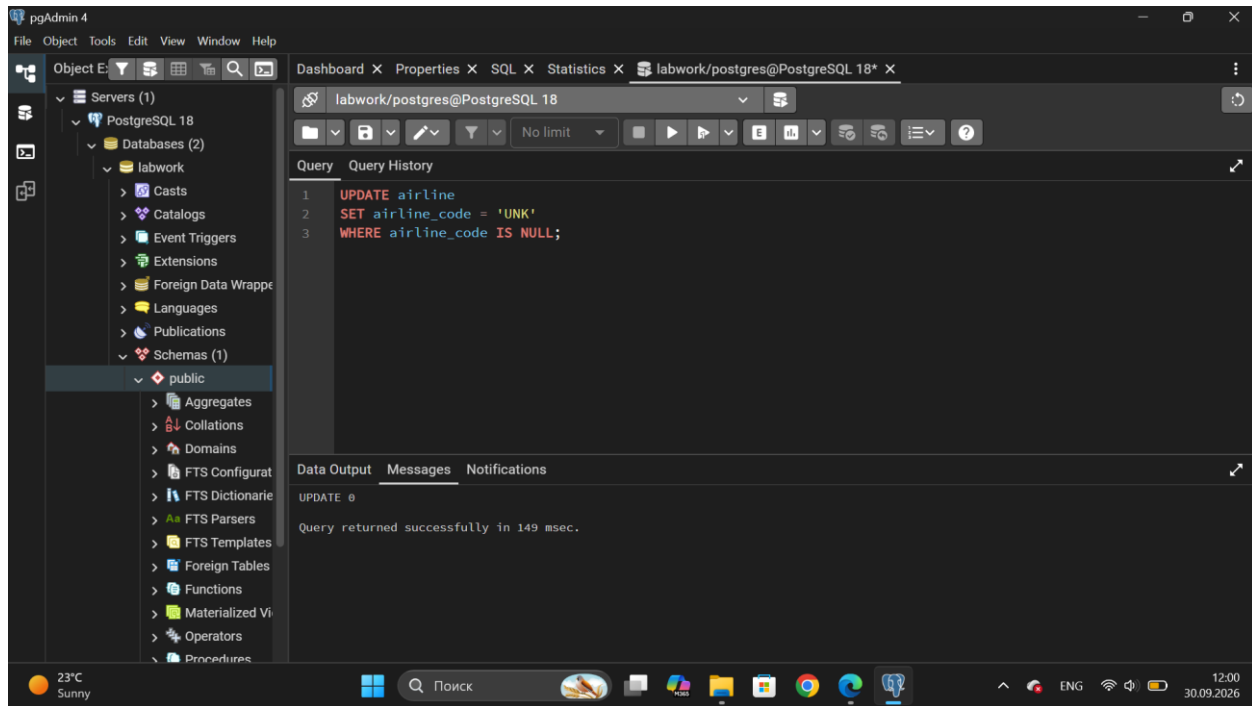
```
1 DELETE FROM booking
2 WHERE ticket_price < 10000;
```

The 'Messages' tab is active, displaying the execution status of the query:

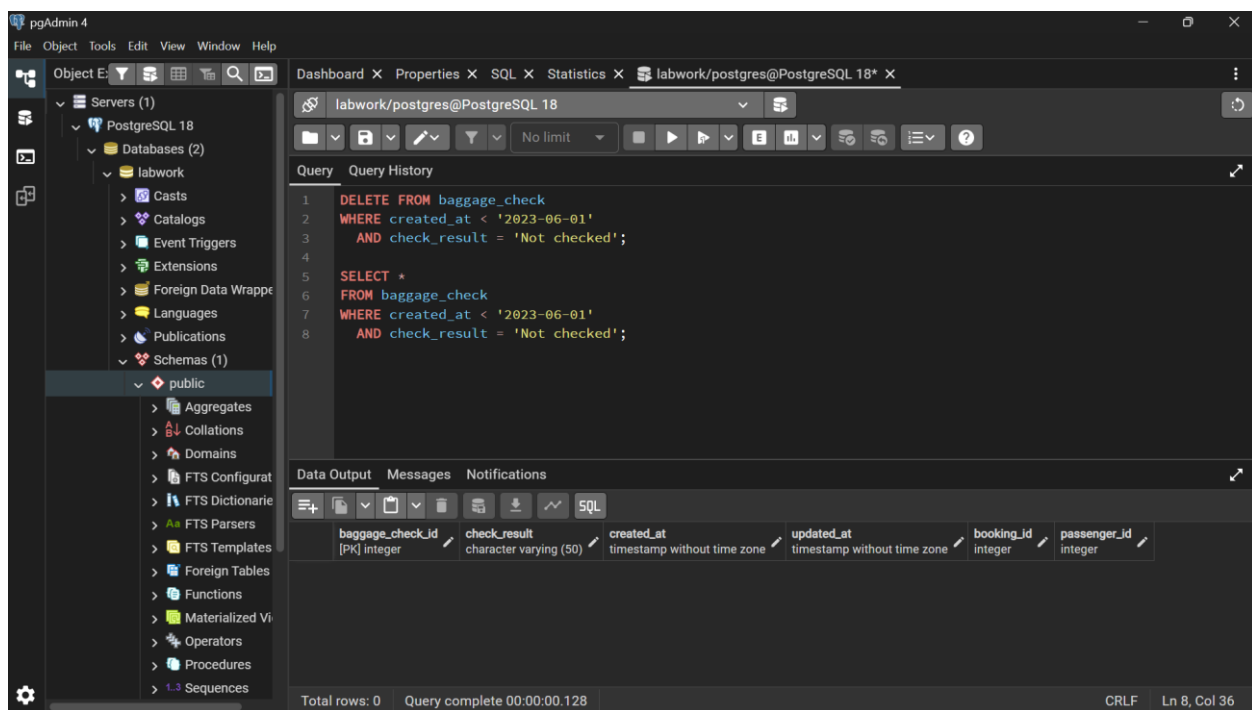
```
DELETE 0
Query returned successfully in 72 msec.
```

A green status bar at the bottom right indicates 'Query returned successfully in 72 msec.' The status bar at the bottom also shows 'Total rows: 0' and 'Query complete 00:00:00.072'.

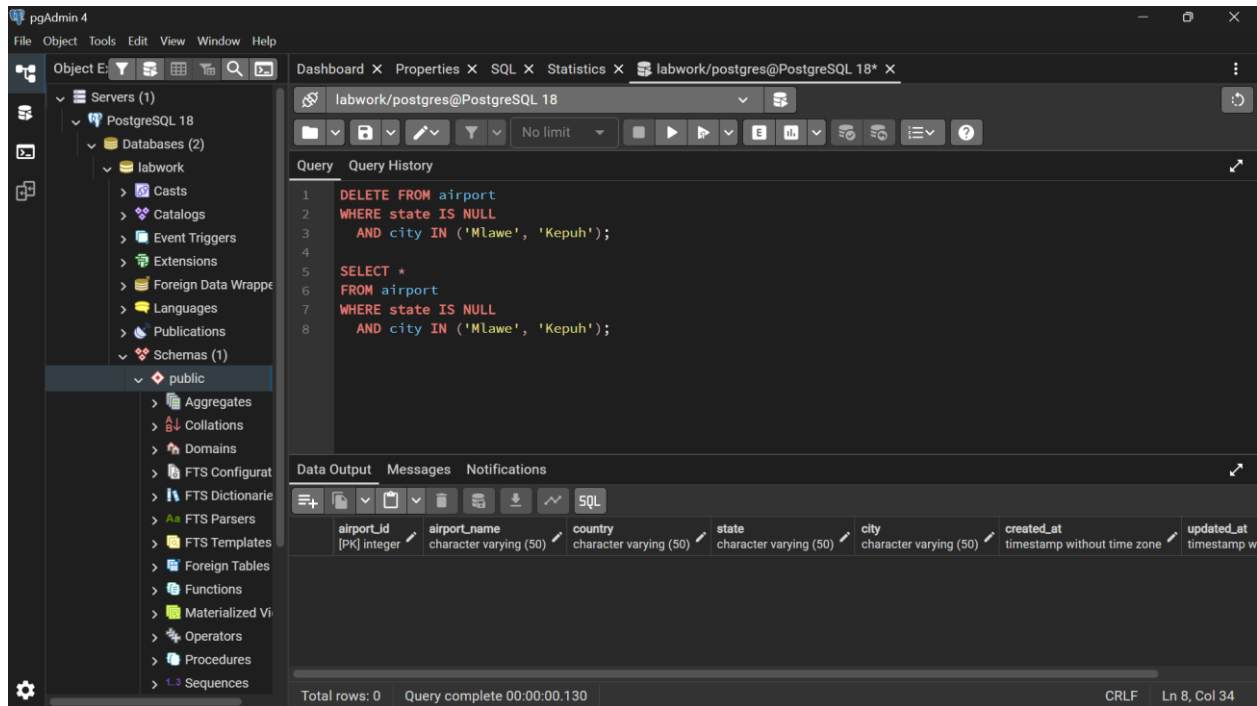
8. Set a default code 'UNK' (Unknown) for any airline in the airline table where airline_code is missing (NULL).



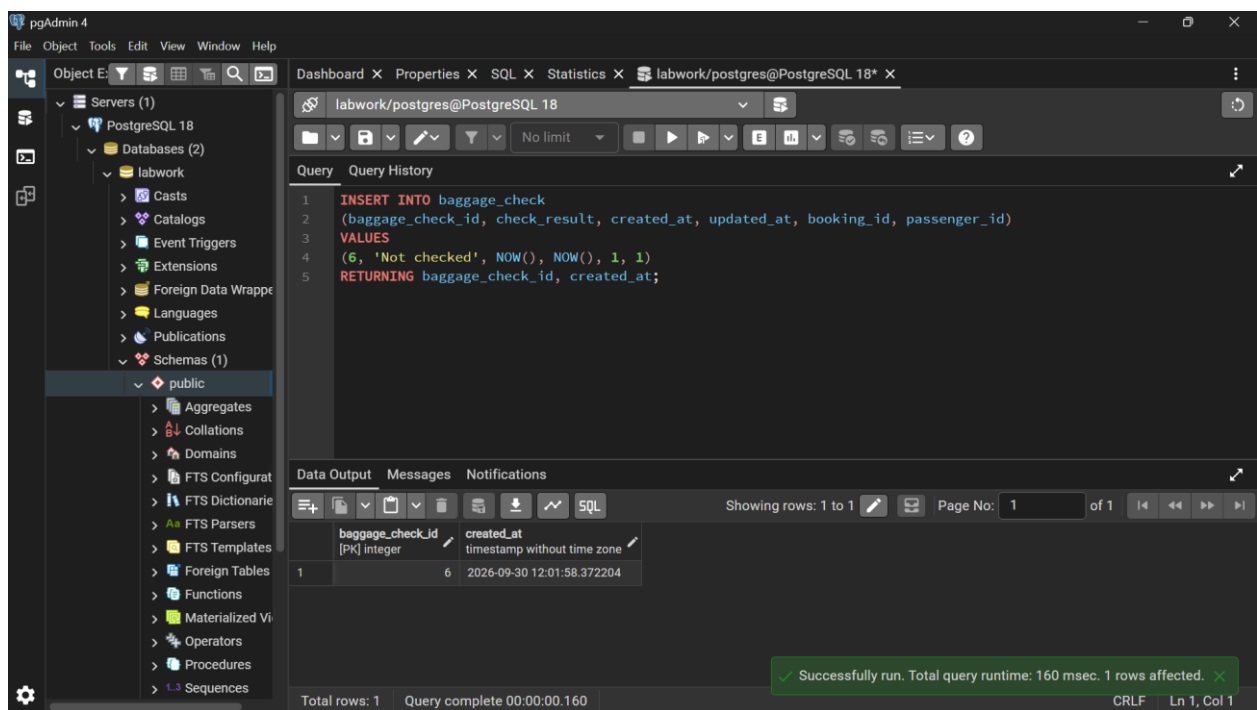
9. Delete all baggage check records created before June 1, 2023, whose result is still 'Not checked'.



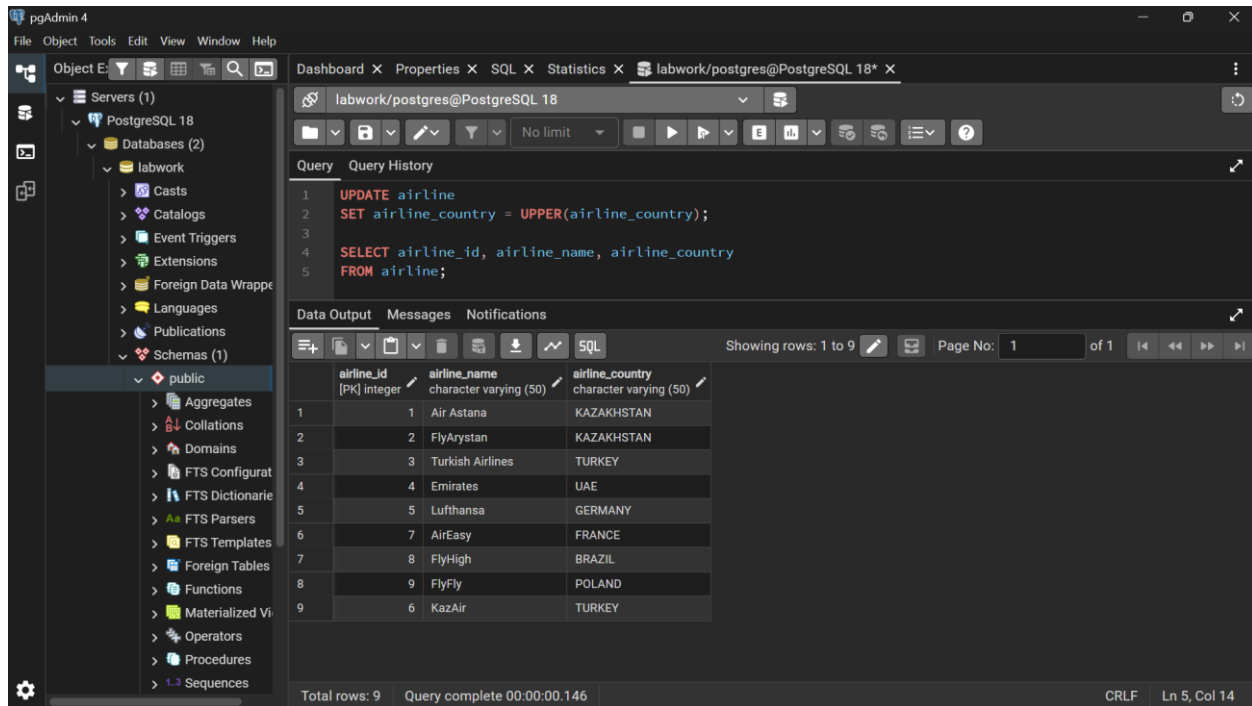
10. Delete all entries from the airport table where state is missing (NULL) and the city is either 'Mlawe' or 'Kepuh'.



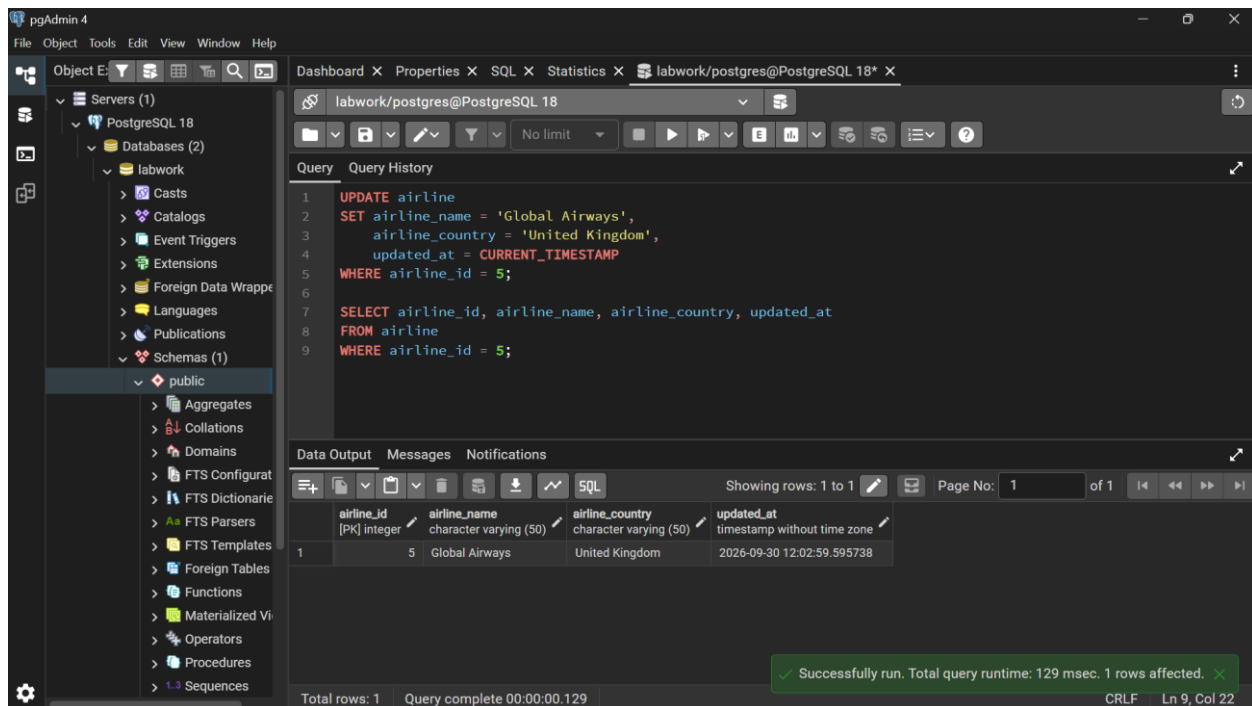
11. Insert a new unchecked baggage inspection record and immediately return its newly generated baggage_check_id and created_at timestamp.



12. Convert all country names in the airline table to uppercase to ensure data consistency.



- Update both the name and country for the airline with `airline_id = 5`. Set the name to 'Global Airways' and the country to 'United Kingdom', while updating `update_at` to the current date.



- Set the state field to 'Capital District' for airports located in specific cities: 'Astana', 'London', and 'Tokyo'

pgAdmin 4

File Object Tools Edit View Window Help

Dashboard x Properties x SQL x Statistics x labwork/postgres@PostgreSQL 18* x

labwork/postgres@PostgreSQL 18

Query Query History

```
1 UPDATE airport
2 SET state = 'Capital District'
3 WHERE city IN ('Astana', 'London', 'Tokyo');
4
5 SELECT airport_id, airport_name, city, state
6 FROM airport
7 WHERE city IN ('Astana', 'London', 'Tokyo');
```

Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1 of 1

airport_id [PK] integer	airport_name character varying (50)	city character varying (50)	state character varying (50)
1	2 Astana International Airp...	Astana	Capital District

Successfully run. Total query runtime: 187 msec. 1 rows affected.

Total rows: 1 Query complete 00:00:00.187 CRLF Ln 7, Col 45

15. Mark all baggage check records (baggage_check) created in March 2024 with a status of 'Not checked' as completed ('Checked').

pgAdmin 4

File Object Tools Edit View Window Help

Dashboard x Properties x SQL x Statistics x labwork/postgres@PostgreSQL 18* x

labwork/postgres@PostgreSQL 18

Query Query History

```
1 UPDATE baggage_check
2 SET check_result = 'Checked'
3 WHERE created_at >= '2024-03-01'
4 AND created_at < '2024-04-01'
5 AND check_result = 'Not checked';
6
7 SELECT *
8 FROM baggage_check
9 WHERE created_at >= '2024-03-01'
10 AND created_at < '2024-04-01';
```

Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1 of 1

baggage_check_id [PK] integer	check_result character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone	booking_id integer	passenger_id integer
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Successfully run. Total query runtime: 159 msec. 0 rows affected.

Total rows: 0 Query complete 00:00:00.159 CRLF Ln 10, Col 33